

RAMACHARAN REDDY KASIREDDY

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AI Engineer with 3+ years of experience delivering production ML/GenAI end-to-end—predictive analytics with feature engineering and statistical validation, and governed RAG/chat agents that ground responses in verified data. Proven capability across data pipelines (SQL/PostgreSQL), model development and evaluation, and secure cloud delivery (AWS/Azure) with Terraform and CI/CD for measurable business impact.

SKILLS

AI/ML & LLMs	LangChain, LangGraph, MCP, RAG (hybrid k-NN + BM25), Prompt Engineering, Fine-Tuning, Multi-Agent Systems, ReAct, Tool Calling, Structured Outputs, Guardrails, Langfuse, Semantic Search, Conversational AI
LLM Platforms	Azure OpenAI (GPT-4, Embeddings), AWS Bedrock (Claude, Titan Embeddings), Google Gemini
Deep Learning	PyTorch, TensorFlow, HuggingFace Transformers, LoRA, PEFT, QLoRA, Quantization, Distillation
Programming	Python, SQL, pandas, statsmodels, scikit-learn, Java, FastAPI, Uvicorn, Pydantic, Jinja, Playwright, pytest
Data & APIs	PostgreSQL, Cosmos DB, OpenSearch, SQLAlchemy, RESTful APIs, OpenAPI/Swagger, OAuth 2.0, JWT, MSAL, JWKS, SSE, WebSockets
AWS	Bedrock, Bedrock Guardrails, OpenSearch, EKS, ECR, Lambda, SQS, S3, Transcribe, DynamoDB, CodePipeline, KMS, IAM (IRSA)
Azure	OpenAI Service, Cosmos DB, ACR, Azure Container Instance, Azure Communication Services, Key Vault, SharePoint, Microsoft Graph, Managed Identity, Logic Apps
DevOps & Tools	Docker, Kubernetes, Helm, Terraform, Airflow, Databricks, Spark, MLflow, Git, GitHub Actions, Jenkins, GitLab CI/CD, GitHub Copilot, Cursor, Claude Code

WORK EXPERIENCE

Feuji INC AI Engineer | Aug 2024 – Present

Ask the Prospect (ATP) – TriNet

Python, FastAPI, LangGraph, AWS Bedrock, Bedrock Guardrails, OpenSearch, Aurora, Langfuse, Airflow, EKS

TriNet's sales reps typically take 2 years to reach full proficiency. The client asked for a searchable training knowledge base – I expanded the scope to a multi-mode AI agent combining RAG Q&A, persona-based role-play coaching, sales analytics, and gamified learning to compress that ramp-up across the entire sales organization.

- **Built a multi-mode LangGraph agent** powering 4 capabilities – RAG chatbot (dual-index: training docs + Gong transcripts), persona-based sales role-play with AI coaching, analytics chat, and automated Jeopardy question generation.
- **Enabled memory continuity** via LangGraph PG checkpointer and store for short- and long-term memory.
- **Engineered a sales analytics pipeline** – LLM enriches each Gong transcript with derived columns (win/loss, objections, competition mentions), then generates OpenSearch queries from natural language for aggregate analysis.
- **Enforced PII protection with AWS Bedrock Guardrails** – configurable filters that detect and redact sensitive data (names, emails, SSNs) in both inputs and outputs before reaching the client.
- **Integrated Langfuse for observability and cost monitoring** – LLM-as-judge evaluations for quality scoring, trace-level debugging, latency monitoring, and per-call cost tracking in production.
- **Served as solution architect and client-facing lead**, owning end-to-end technical vision across 4 microservices and 2 Airflow-orchestrated data pipelines on EKS.
- **Recognized by TriNet's CTO, Executive Director, and VP** for expanding scope beyond original requirements and delivering measurable impact to the sales organization.

Customer Analytics – PLZ Corp

Python, Microsoft Copilot Studio, Azure (OpenAI, PostgreSQL, Communication Services, Container Instance & Registry, Logic Apps, Key Vault)

PLZ needed daily, verified financial intelligence. I built Customer Analytics end-to-end: a daily pipeline that ingests financial statements, regulatory filings, and corporate news, processes 10-K/10-Q filings into RAG-ready PDF/text, stores normalized results in Azure PostgreSQL, and saves documents to SharePoint so Copilot Studio can auto-index and power a Teams Finance Agent.

- **Built the daily ingestion/processing pipeline in Python:** pull financial statements, fetch 10-K/10-Q filings, ingest corporate news, dedupe with Azure OpenAI embeddings (cosine similarity ≥ 0.70), and persist metrics to Azure PostgreSQL.
- **Made the job incremental and idempotent** using per-company sync state, keeping the DB, SharePoint documents, and email digests consistent day over day.
- **Provisioned the Azure backend with Terraform IaC** (Key Vault, PostgreSQL, ACR/ACI, Logic Apps, ACS) so the pipeline runs daily with secretless Managed Identity.
- **Stored 10-K/10-Q filings in SharePoint for Copilot auto-indexing** by converting filing HTML to PDF/text and uploading via Microsoft Graph.
- **Configured the Finance Agent in Microsoft Teams** using Copilot Studio, grounded in the SharePoint-indexed filings and daily news corpus.
- **Sent the daily email blast** from the same scheduled run via Azure Communication Services to the distribution list.
- **Delivered Power BI reporting** with a live report tracker (availability + next estimate) and income-statement trend dashboards (revenue, gross profit/margin, EBITDA, operating/net income).
- **Recognized by PLZ Corp's Director and CIO** for exceeding expectations with faster, verified decision support.

AI-Powered IVR Automation – Cotiviti

Java, Python, LangGraph, AWS Bedrock, AWS Transcribe, DynamoDB, Lambda, SQS, KMS, PostgreSQL, Twilio

Cotiviti's Medicare IVR verification ran on legacy Jabber VMs, requiring a human to manually review and update navigation paths every morning. The team migrated to a fully serverless, horizontally scalable AWS architecture and introduced an AI agent that autonomously discovers the correct IVR path daily, eliminating the manual review step entirely.

- **Implemented a LangGraph decision-tree workflow** for Medicare eligibility flows (RRE, NPI, Contractor) using entitlement checks on each transcript to route IVR steps or cut calls.
- **Built a supervised fine-tuning dataset on AWS Bedrock** by generating ordered Q/A scripts for all three flows and creating 100 variations per Q/A pair; used flow-based prompting with the fine-tuned model to select the next step.
- **Automated the call-processing loop** using Twilio recordings + AWS Transcribe, routing per Medicare number via LangGraph, and persisting transcripts and outcomes in DynamoDB/PostgreSQL for 200,000+ daily calls.
- **Ensured release quality and cost savings** with a mock Medicare IVR simulator for regression, KMS-encrypted PII handling via CodePipeline, and results including 730+ engineer-hours saved annually and 60% infra cost reduction.

ReQon – Automated Testing Tool

Python, LangChain, LangGraph, Google Gemini, Playwright MCP, Aurora PostgreSQL, S3, Docker, pytest

Feuji is a QA-heavy company where test engineers were spending the bulk of their time writing and maintaining low-level test cases. I led an internal initiative to build an AI platform that auto-discovers application structure and generates test cases and regression scripts from a single browser recording, freeing engineers to focus on high-value exploratory testing.

- **Engineered a LangGraph agent with Playwright MCP** that replays a recorded browser flow, navigates every page, and extracts locators, functional elements, and behavioral data into S3 and Aurora PostgreSQL.
- **Automated the full QA authoring workflow** – from captured browser behavior the agent generates test cases, executes them via Playwright MCP, and outputs verified pass/fail results, eliminating manual test writing.
- **Built a code generation engine** using Jinja templates that converts page metadata into pytest-compatible regression scripts with Page Object Model classes for every discovered page.
- **Designed and dockerized 3 microservices** – Recorder (Playwright codegen), MCP Service (AI agent + test generation), and Execution Service (isolated runner) – saving 100+ team-hours weekly and cutting QA costs by \$230K/year.

University at Buffalo

AI Research Assistant | Jan 2024 – May 2024

Energy-Efficient LLM Inference and Carbon-Aware Optimization

Python, PyTorch, TensorFlow, Hugging Face, NumPy, Pandas

- **Benchmarked parameter-efficient fine-tuning (PEFT) techniques** – LoRA, knowledge distillation, and post-training quantization (4-bit/8-bit) – on GPT-2, measuring GPU energy consumption and carbon emissions across task configurations.
- **Achieved 19.8% CO2 reduction** with only 6% perplexity increase using 8-bit quantization, demonstrating significant energy savings with minimal quality degradation.
- **Demonstrated 45.2% emissions reduction** by combining distillation with quantization, mapping the accuracy-efficiency trade-off curve for carbon-aware deployment.

Flable AI

Data Scientist | Mar 2023 – Dec 2023

Flable Data Insights & Forecasting

Python, Azure, Databricks, Spark, MLflow, statsmodels, PostgreSQL, Jenkins

- **Built client analytics pipelines** on Azure using Databricks/Spark to transform clickstream and funnel events into model-ready datasets stored in PostgreSQL.
- **Engineered predictive features** for click-rate, success-rate, and funnel progression using derived engagement metrics, and applied ANOVA to validate which segments/feature groups differed significantly.
- **Developed forecasting and trend models** using boosting models (e.g., XGBoost) to estimate probabilities of user progression over time and identify key drivers.
- **Operationalized daily scoring** by turning Jupyter experiments into repeatable Python batch jobs, scheduled via Jenkins to refresh outputs for clients.
- **Tracked model iterations with MLflow** to compare runs, manage metrics, and support repeatable updates.

Flable Digital Assistant

Python, LangChain, OpenAI, PostgreSQL, FastAPI, Docker, Jenkins, GitLab

- **Built a LangChain conversational AI with OpenAI** that classifies user intent via transformer embeddings, maps queries to PostgreSQL data categories, and generates natural-language responses grounded in real results.
- **Developed an on-the-fly visualization engine** that dynamically queries PostgreSQL, renders charts, and delivers them inline via Telegram – giving non-technical users instant access to analytics.
- **Deployed the full stack** (FastAPI backend + Telegram bot) using Docker with GitLab CI/CD and Jenkins pipelines, reducing client's weekly analysis time by 60%.

OPEN SOURCE

langgraph-checkpoint-cosmos |  github.com/LangModule/checkpoint-cosmos

- Published a LangGraph checkpoint saver for Azure Cosmos DB with sync/async support and keyless DefaultAzureCredential authentication.

EDUCATION

University at Buffalo, The State University of New York

Buffalo, New York

Master of Science, Computer Science, CGPA: 3.7/4

Aug 2023 – Dec 2024

Coursework: Data Intensive Computing, Computer Vision, Machine Learning, Deep Learning